AG News Classification using Machine Learning and NLP

Automatically classifying news articles into World, Sports, Business, and Science/Technology categories using machine learning techniques.

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Project Overview

The Challenge

With massive digital news content generated daily, manual classification has become inefficient and time-consuming. The system addresses this by automatically analyzing textual content and predicting appropriate categories.

The Solution

An intelligent text classification system using Logistic Regression and TF-IDF vectorization, enhanced with a Streamlit web interface for real-time predictions.

Key Objectives

Automated Classification

Develop system for automatic news article categorization

NLP Techniques

Apply preprocessing and vectorization methods

Model Training

Build and train Logistic Regression classifier

Web Interface

Create Streamlit app for real-time predictions

Dataset & Methodology

01

AG News Dataset

4 balanced categories: World, Sports, Business, Science/Technology

02

Data Preprocessing

Lowercasing, punctuation removal, tokenization, and text cleaning

03

TF-IDF Vectorization

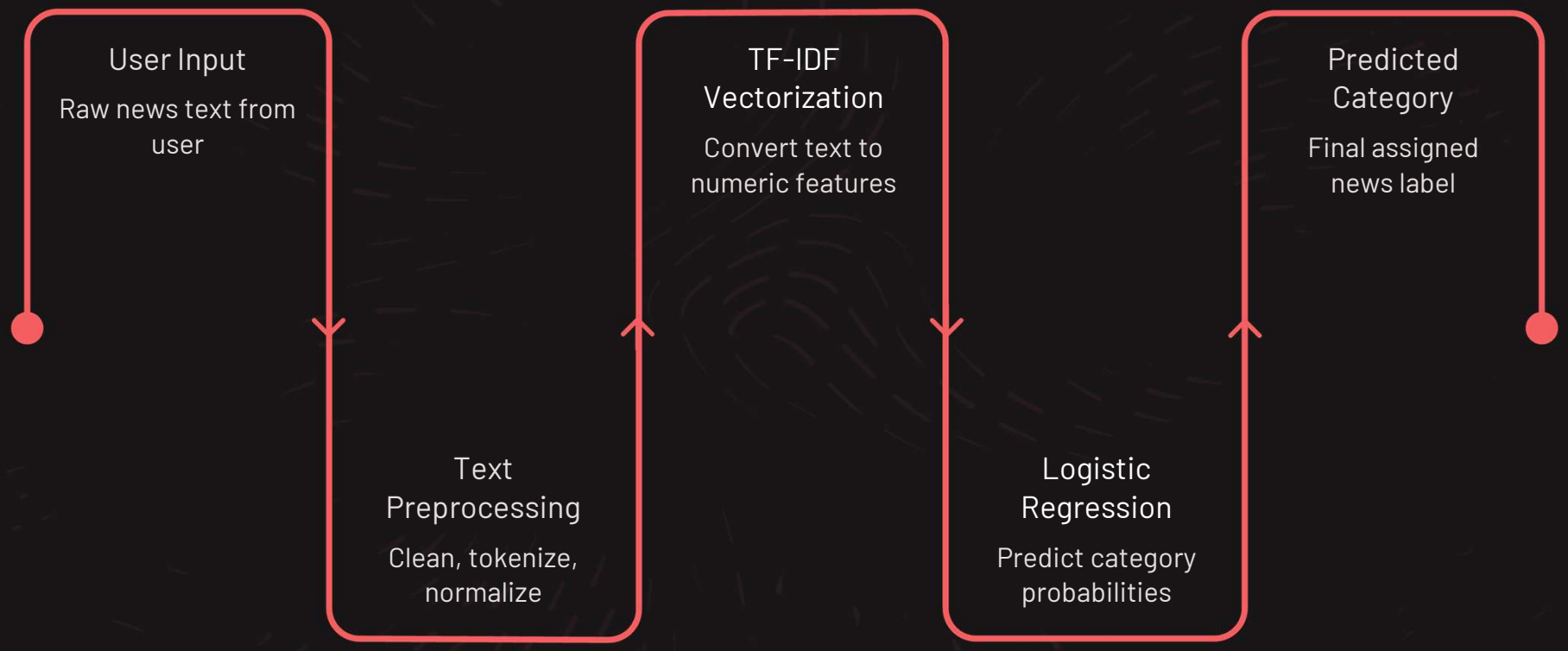
Convert text to numerical format for model processing

04

Model Training

Logistic Regression trained on labeled training data

System Architecture



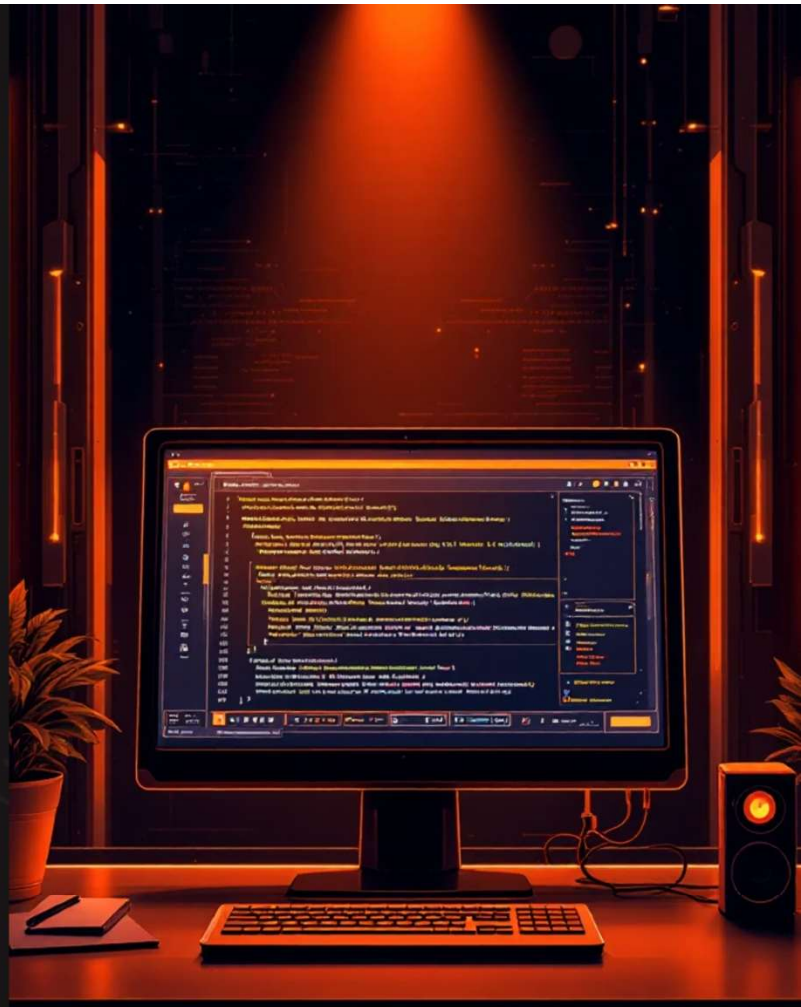
Implementation Details

Python Libraries Used

- Pandas for data manipulation
- NumPy for numerical operations
- Scikit-learn for ML and TF-IDF
- Streamlit for web interface
- Joblib for model persistence

Model Characteristics

Logistic Regression chosen for efficiency with text data, fast training/prediction, and interpretability. Model saved as News_Classificatiom_Model_LR. joblib for deployment.



Performance Metrics

1

Accuracy

Percentage of correctly classified news articles

2

Precision

Correct positive predictions for each category

3

Recall

Ability to identify all relevant instances

4

F1 Score

Harmonic mean balancing precision and recall

Model demonstrates stable and reliable performance on test dataset with balanced class distribution.

Streamlit Web Application

Input Interface

Text area for entering news articles

Classification

Button triggers prediction process

Output Display

Shows predicted category instantly

Applications & Impact

News Websites

Automatic article categorization

Search Engines

Improved indexing and search results

Recommendation Systems

Personalized news suggestions

Social Media

Content filtering and organization

Future Enhancements

Advanced Models

Random Forest, XGBoost, SVM

Deep Learning

LSTM, BERT transformers

Multilingual

Support multiple languages

Real-Time API

Live news integration

This project establishes a strong foundation for building more advanced, scalable text classification applications in the future.